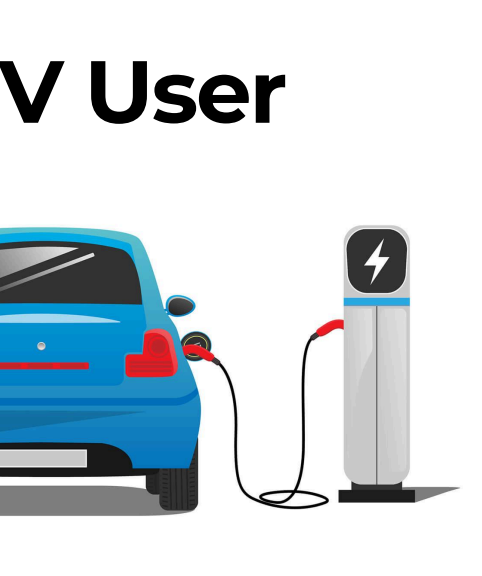
A UX Case Study on EV User Experience

Design Research

Ayaan Ahmad



Problem Statement

Even as electric vehicles become more affordable and desirable, their adoption is still constrained by unreliable, inaccessible, and slow charging infrastructure. Unlike traditional petrol/diesel vehicles, where drivers can confidently plan trips knowing that fuel stations are widely available, EV owners face uncertainty about charger availability, location, and compatibility.

Primary Research - Google Form

36 responses

Majority	Majority	Majority
Struggle to find charging stations.	**Want live availability in locators.**	**Find charging speed too slow.**

Moderate	Moderate	6.6/10
Don't know how to monitor battery health.	**Face unreliable battery range.**	**Average rating of a user's overall EV experience.**

Insights from Open Responses

Range concerns: Users consistently wish for better battery range and performance.
Battery issues & warranty gaps: Frustration with battery degradation, swelling, and lack of warranty coverage.
Service accessibility: Lack of nearby service centers makes maintenance and support difficult.
Charging speed: Users want faster charging options for convenience.
Build quality & reliability: Complaints about EVs switching off at low charge or glitches in rain show need for improved durability feedback loops.

Interview Insights

Battery & Range

- Unreliable range: Displayed range often doesn't match real-world performance.
- Battery drains fast on certain trips despite showing high range.
- Even "fast charging" feels slow compared to fueling a petrol vehicle.
- Reduced range as batteries age.
- Swappable battery suggested as a convenience.

Charging Infrastructure

- Rely heavily on home charging due to limited public options.
- Lack of public charging stations creates range anxiety and travel restrictions.
- Arrive at a station only to find chargers occupied, out of service, or incompatible.
- Waiting for charging causes frustration and delays in daily routines.

Digital & App Experience

- Navigation needs integration: Users want charging station navigation directly in the vehicle/app.
- Desired app features: Location tracking, remote vehicle lock, charging reminders.
- Battery health metric seen as "useless" unless it translates into practical insights.

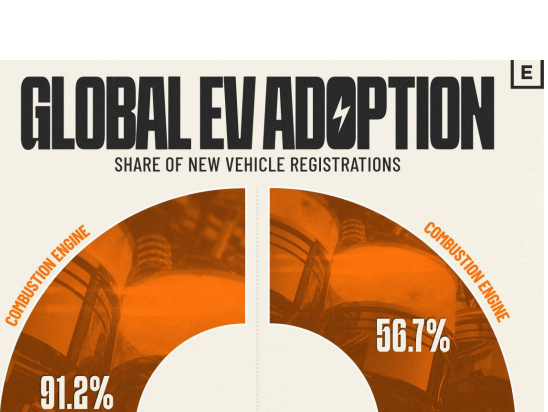
Reliability & Service

- Frequent mechanical/electronic issues requiring service interventions.
- Ola's after-sales service is poor, with limited service centers and no structured plans.
- Hazard sign issue creates repeated restarts.

Vehicle Design & Usability

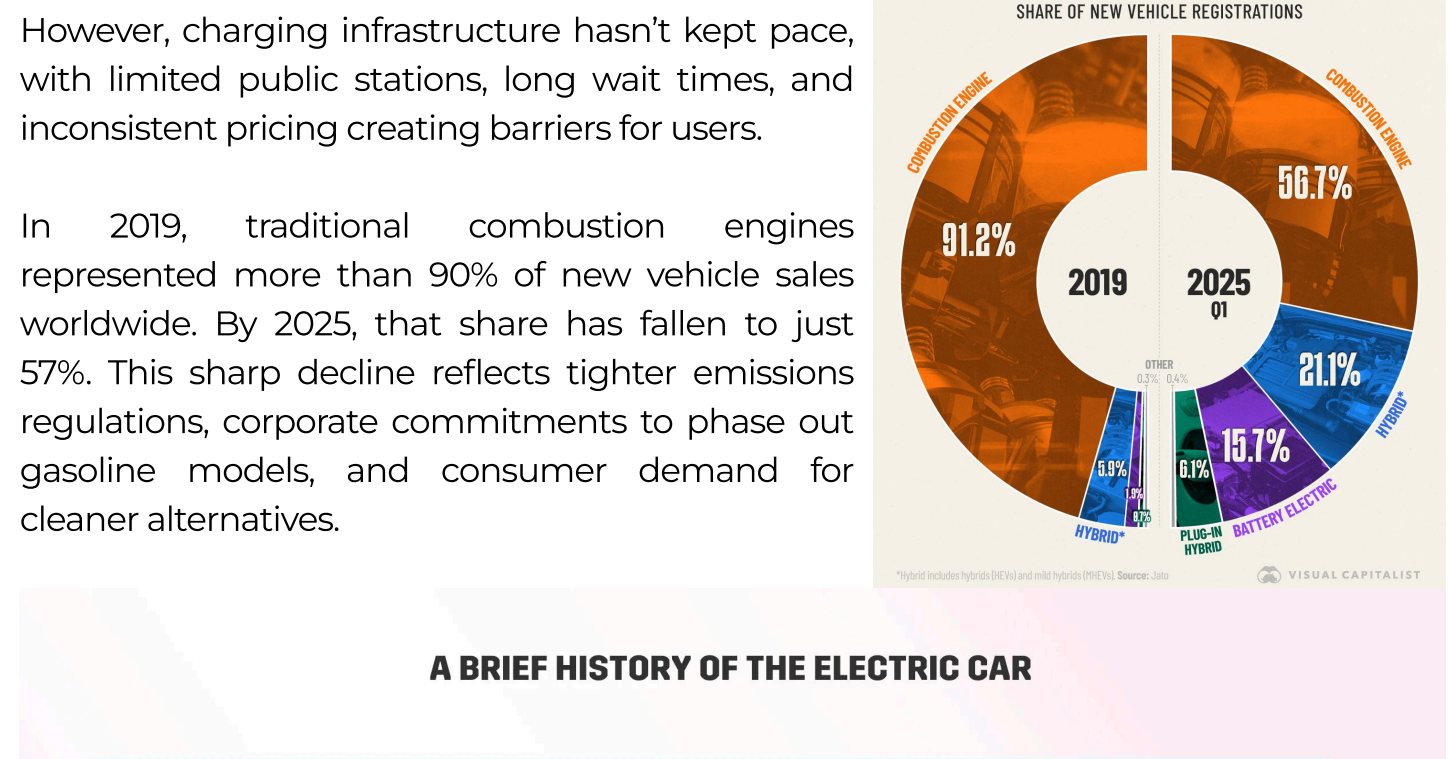
- Boot space is necessary.
- Build quality and styling matter.
- EV scooters feel lighter and smoother than petrol ones.
- Touchscreens can hinder usability when they lag or freeze.

Secondary Research



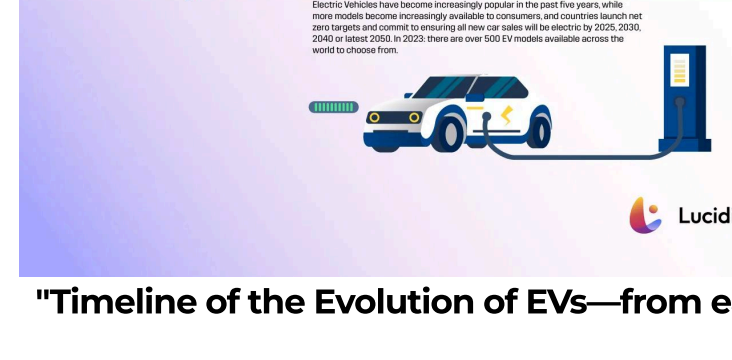
EV adoption is rapidly increasing, with nearly **19% of new car sales being electric in 2025.**
However, charging infrastructure hasn't kept pace, with limited public stations, long wait times, and inconsistent pricing creating barriers for users.

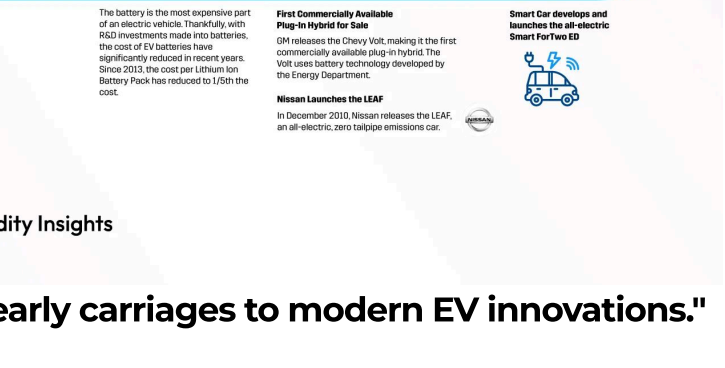
In 2019, traditional combustion engines represented more than 90% of new vehicle sales worldwide. By 2025, that share has fallen to just 57%. This sharp decline reflects tighter emissions regulations, corporate commitments to phase out gasoline models, and consumer demand for cleaner alternatives.



"Timeline of the Evolution of EVs—from early carriages to modern EV innovations."

The first recorded attempts at developing an electric vehicle can be traced back to 1828, when horses and buggies were still the main mode of transportation.
The first hybrid electric vehicle was developed by Porsche in 1901. But cheap oil prices, and affordable gas-powered vehicles like Ford's famous Model T make electric vehicles a distant memory for much of the 1900's.
Soaring gas prices in 1968 bring back some interest, and finally the 1970's bring us the first commercially available EVs.
But it wasn't until the launch of the Toyota Prius, the first mass-produced hybrid in 1997, that the world considers beginning the transition to electric powered vehicles.
Then, Tesla motors comes on the scene in 2006 and is incites a revolution.



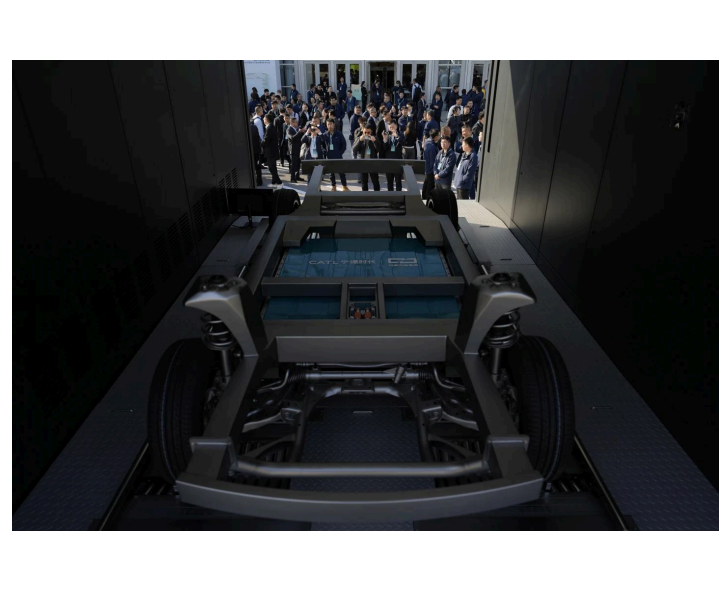


Here's two images showcasing the three levels of EV charging (Level 1, Level 2, Level 3/ DC Fast Charger) It effectively illustrates the evolution from slow home-based charging to fast, high-voltage infrastructure.

CASE STUDIES

1. China's getting a big electric car battery swapping boost in 2025
In 2025, China is significantly expanding its EV battery swapping infrastructure, led by CATL and NIO. This model allows drivers to swap depleted batteries for fully charged ones in minutes, reducing charging time and battery degradation. The system works well in China due to standardized vehicle platforms and strong government support. However, global adoption faces barriers like lack of standardization, infrastructure cost, and limited compatibility across EV brands.

CATL 宁德时代
力换电



2. Scaling India's EV Charging Network
India's EV charging infrastructure has grown 5 times in the last 3 years, reaching over 26,367 public charging stations by 2025. Despite this rapid expansion, the country still has only 1 charger for every 235 EVs, indicating a major accessibility gap. The highest concentration of stations is in cities like Bengaluru and Delhi. Government initiatives like FAME III are pushing growth, but rural and highway coverage remains limited.

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FAME-India
(National Mission on Electric Mobility)

FAME-III: New Milestone for Electric Mobility in India

FAME I (2015-2020)

- Launched with an initial outlay of Rs. 900 crore.
- Focused on providing demand incentives to EV buyers.
- Supported 278,000 pure EVs with a total incentive of Rs. 343 crore.

FAME II (2020-2024)

- Increased outlay of Rs. 10,000 crore, later revised to Rs. 11,500 crore to meet growing demand.
- Supported over 1.6 million vehicles and witnessed an increase in the number of EV players from 124 to 731 between FY15 and FY24.
- Led to a significant rise in EV sales, from less than 7,000 units in FY15 to 1.5 million units in FY24, making up 6.8% of total automobile sales.

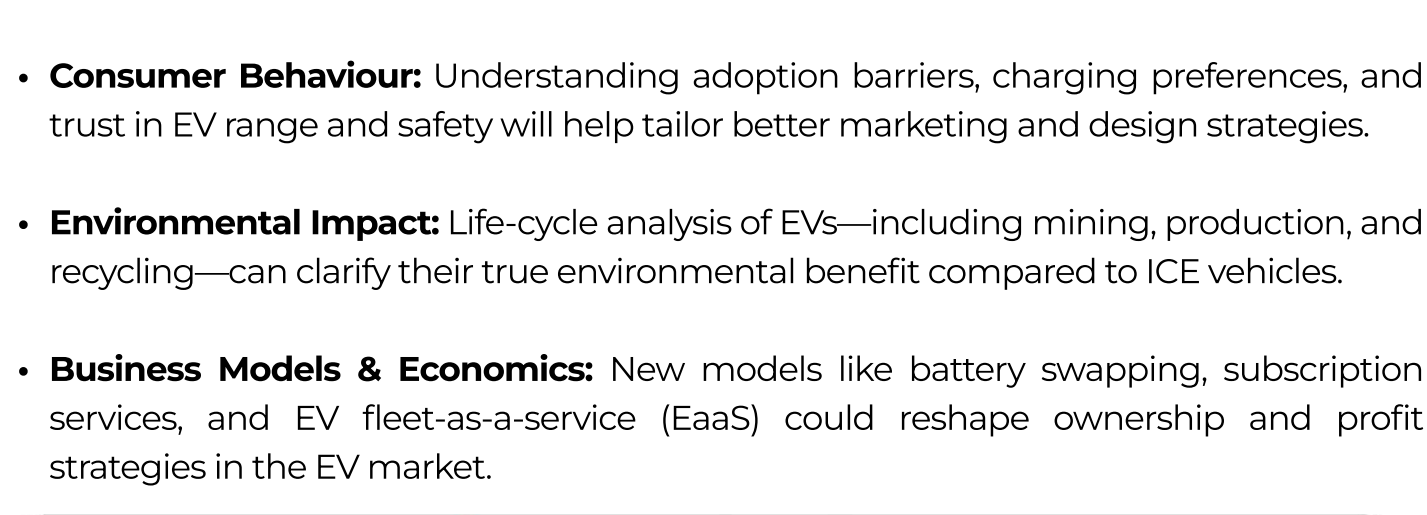
FAME-III (Upcoming Phase)

- Set to launch in the next two months, FAME-III will continue India's commitment to electric mobility, with a focus on expanding EV adoption across various vehicle segments, including heavy-duty trucks.

3. Ireland's 'first-of-its-kind' home-charging EV pilot rolled out
In June 2025, Ireland launched a first-of-its-kind pilot offering free EasyGo home EV chargers to selected households without off-street parking. The initiative aims to solve the home charging gap in urban areas, where many residents can't install private chargers. It's part of a broader push to accelerate EV adoption and reduce dependence on public charging. This model could serve as a blueprint for similar programs in other countries facing urban infrastructure challenges.

FUTURE RESEARCH DIRECTIONS

- **Battery Technology:** Research can explore solid-state batteries, faster charging chemistries, and recyclable materials to increase range, safety, and sustainability while reducing costs.
- **Charging Technology:** Advancements in ultra-fast charging, wireless charging, and vehicle-to-grid (V2G) systems could improve charging convenience and grid stability.
- **Grid Integration:** Studies on smart grid management, dynamic load balancing, and EV demand forecasting are needed to ensure the power grid can support large-scale EV adoption.
- **Consumer Behaviour:** Understanding adoption barriers, charging preferences, and trust in EV range and safety will help tailor better marketing and design strategies.
- **Environmental Impact:** Life-cycle analysis of EVs—including mining, production, and recycling—can clarify their true environmental benefit compared to ICE vehicles.
- **Business Models & Economics:** New EV models like battery swapping, subscription services, and EV fleet-as-a-service (EaaS) could reshape ownership and profit strategies in the EV market.



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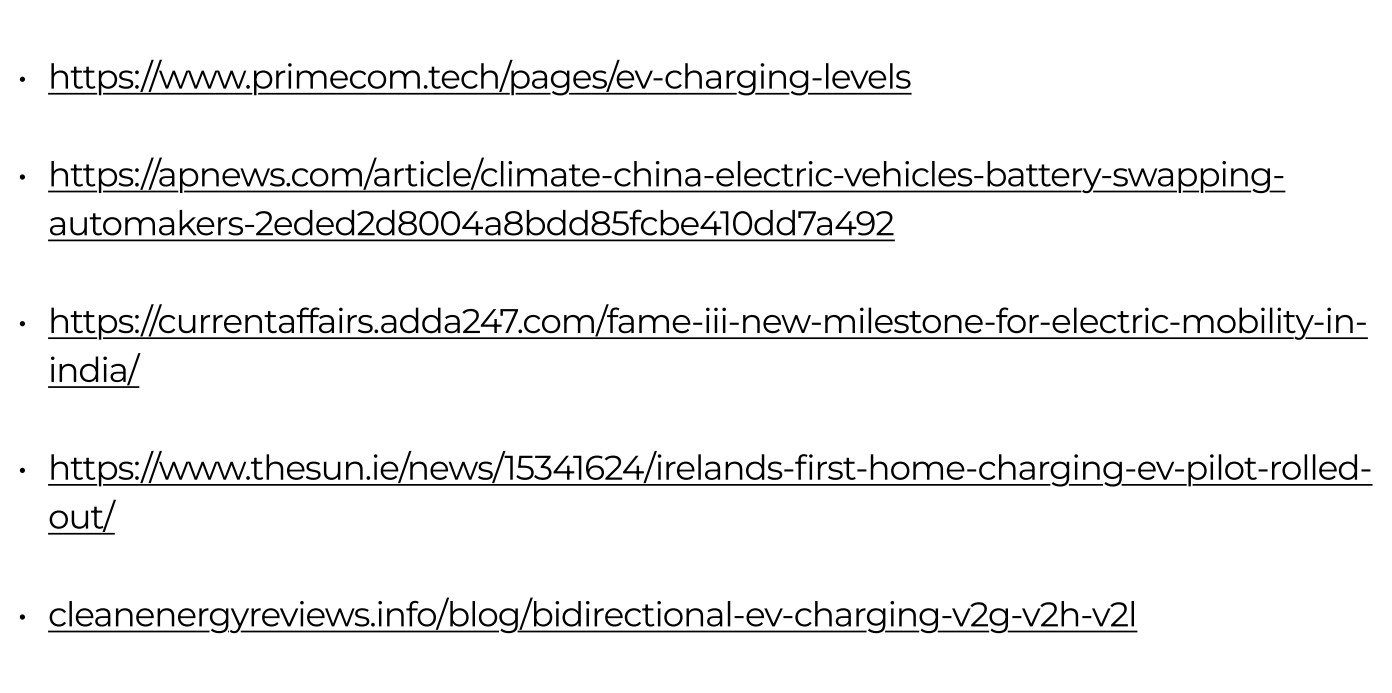
User Persona



Age
28

Profession
IT Professional

Background
Arjun Rao, uses his EV scooter for daily city commutes. While he values its low cost and eco-friendliness, he faces issues like limited charging options, slow charging, and lack of service support. He seeks a more seamless EV experience with smarter charging and battery management tools.

Arjun Rao

Goals & Motivation

- Wants a hassle-free daily commute with his EV.
- Aims to save costs on fuel while being eco-friendly.
- Seeks reliable charging support without wasting time.

Pain Points

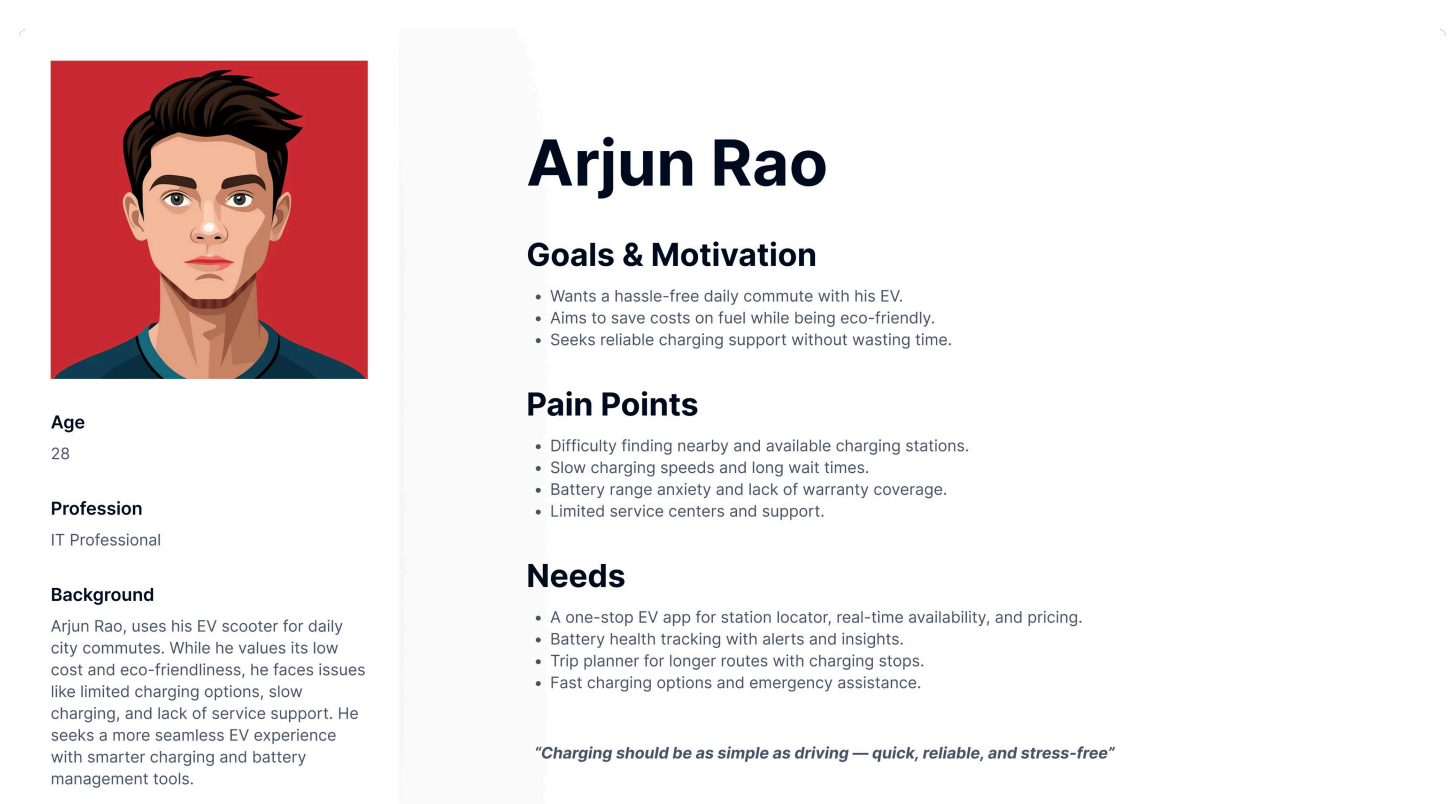
- Difficulty finding nearby and available charging stations.
- Slow charging speeds and long wait times.
- Battery range anxiety and lack of warranty coverage.
- Limited service centers and support.

Needs

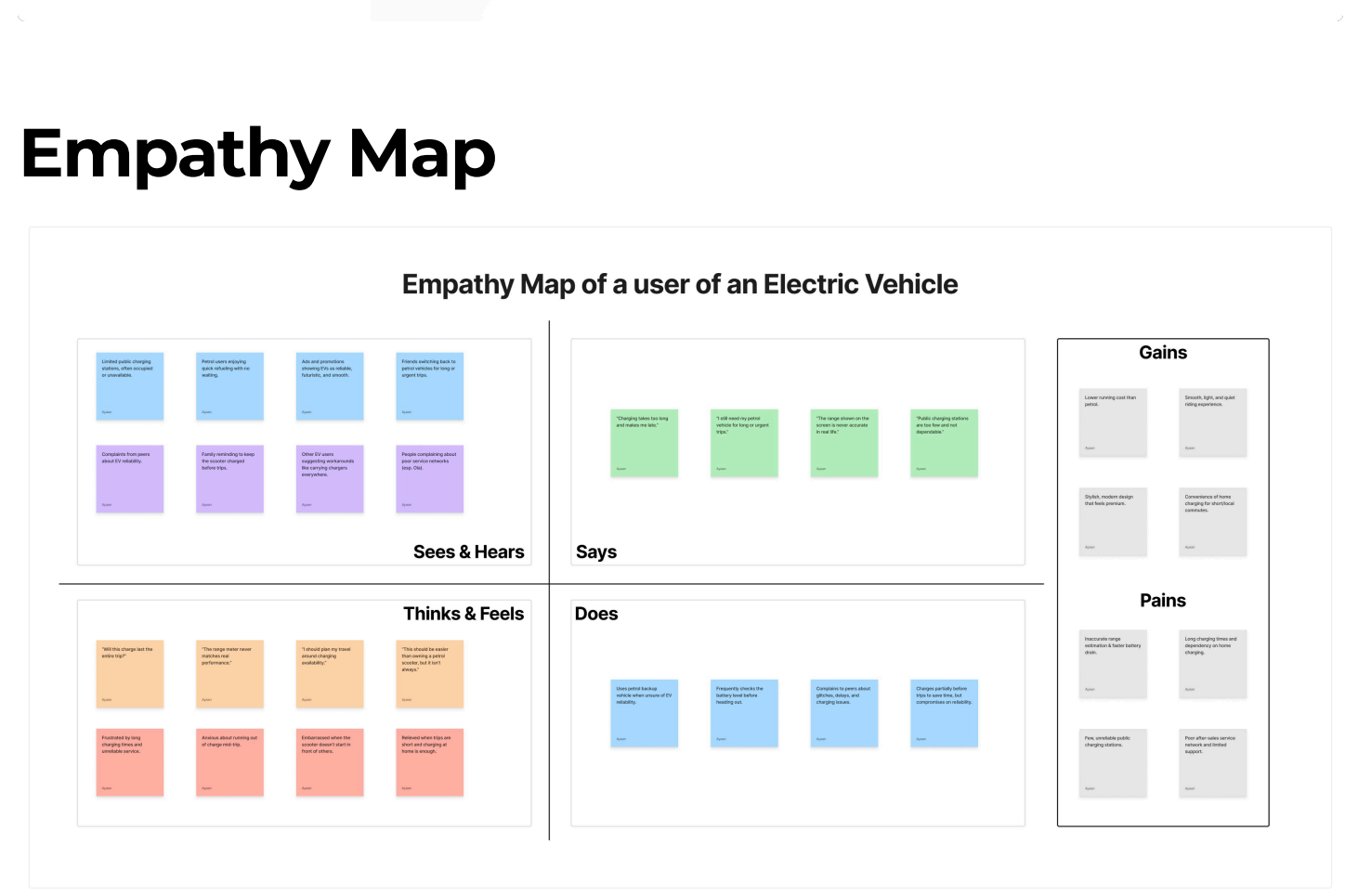
- A one-stop EV app for station locator, real-time availability, and pricing.
- Battery health tracking with alerts and insights.
- Trip planner for longer routes with charging stops.
- Fast charging options and emergency assistance.

"Charging should be as simple as driving — quick, reliable, and stress-free"

Empathy Map



Storyboarding



STORYBOARDING
"FROM ANXIETY TO ASSURANCE" - ARJUN'S EXPERIENCE

FRAME 1
Arjun is getting ready for work. He checks his car's dashboard and sees the battery at 56%.

FRAME 2
On the road, the battery drops faster than expected. Arjun looks at the screen nervously.

FRAME 3
He passes by a petrol pump with people refueling quickly, but sees no EV charging station nearby.

FRAME 4
Arjun opens the EV Companion App on his phone and comes across a lot of helpful features.

FRAME 5
He follows in-car navigation (synced with app) towards a nearby charging station.

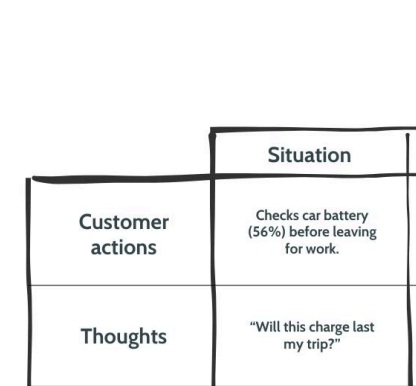
FRAME 6
Arjun charges quickly at the station, relieved, then continues his trip smoothly.

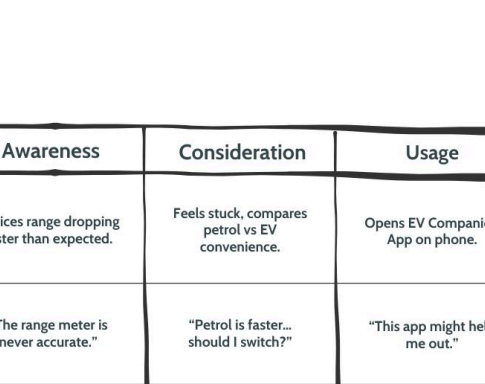
User Journey Mapping

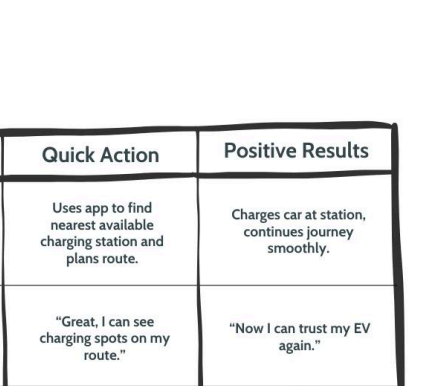
	Situation	Awareness	Consideration	Usage	Quick Action	Positive Results
Customer actions	Checks car battery (56%) before leaving for work.	Notices range dropping faster than expected.	Feels stuck, compares petrol vs EV convenience.	Opens EV Companion App on phone.	Uses app to find nearest available charging station and plans route.	Charges car at station, continues journey smoothly.
Thoughts	"Will this charge last my trip?"	"The range meter is never accurate."	"Petrol is faster... should I switch?"	"This app might help me out."	"Great, I can see charging spots on my route."	"Now I can trust my EV again."
Customer experience	☹️	😔	🤔	🙂	😎	😌
Pain Points	Inaccurate range estimate.	Faster battery drain.	No visible charging infrastructure.	App & car ecosystem often unreliable (general issue).	Fear of delays if station is full, but still happy.	Delays reduced, reliability restored.
Solutions	Real-time range prediction.	Battery health + accurate distance-to-empty insights.	Awareness of charging stations nearby.	Integrated EV app with live charging availability.	Route planning with charging stops + reminders.	Smooth trip, trust in EV improves.

How Might We Statements

- HMW make displayed range more reliable so users can trust their EV on every trip?
- HMW integrate smart navigation with charging station data so trips are planned confidently without range anxiety?
- HMW monitor the overall vehicle battery health?
- HMW translate battery health data into simple, actionable insights that help users make better decisions?



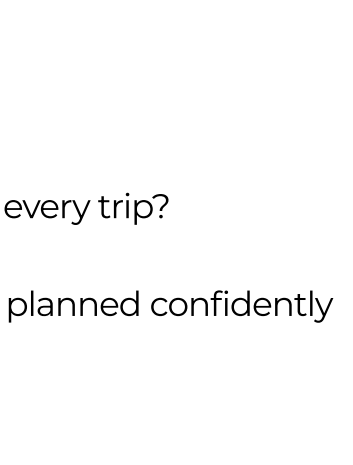




Key fact: "India has 1 charger for every 235 EVs compared to 1 petrol pump for every 3k vehicles."

What it implies: EV infrastructure hasn't yet reached the availability, accessibility, or reliability of petrol pumps and this gap in convenience and trust is what an EV app aims to address.

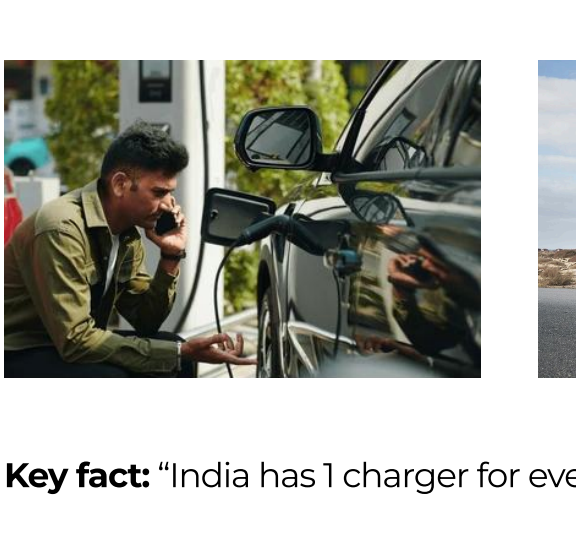
All-in-one Solution Statement



Our solution is a user-centered EV companion app that:

- Builds trust, reduces anxiety, and simplifies the charging journey
- It provides accurate range predictions
- Integrates real-time charging station availability and slot booking
- Offers smart navigation that plans routes with charging stops.
- Predictive charging reminders
- Transparent pricing and payment options
- Actionable battery health insights

By providing all of the features mentioned above, the app allows EV users to enjoy the same freedom and confidence that combustion engine drivers do at petrol stations.



Why an EV companion app is a better option?

- It integrates real-time charging station availability
- Even though the number of current charging stations is small, this app will make each station visible.
- More than just a navigation app, it's a complete vehicle monitoring and intelligence system.
- Clears hesitation, builds trust, reduces anxiety, and simplifies the overall journey of an EV user.

Research? Again?

YES. Understanding and analyzing the typical electric vehicle charging experience was the goal of the thorough study done in the previous phase. It's time to do the same for our competitors such as Tesla, Ather, and others. Since the end result is an EV companion application, let's explore its context in more detail.

COMPETITOR	UPS	DOWNS
Tesla App	Deep vehicle integration (remote charge start/stop, climate preconditioning, Supercharger integration), polished UX for owners.	Phone-key and connectivity glitches reported; premium subscriptions (connectivity) receive user criticism
Ather App	Good baseline ratings; provides vehicle monitoring, trip stats, and Ather network features.	Many users report sync lag, inconsistent pairing, and regressions after software updates. Community forums flag poor app sync as a recurring pain.
Ola Electric App	Integrates OTA features and store/service booking; linked to a broad dealer/service network.	Strong negative sentiment around reliability, app-based unlocking failures, and poor after-sales service. Recent news shows various service complaints.
PlugShare / ChargePoint / Aggregators	Best-in-class for charger discovery, filters, and cross-network support hence essential for trip planning beyond an OEM's network.	Less vehicle-control functionality; sometimes data freshness and station uptime vary by region.

BUT WHAT IF...

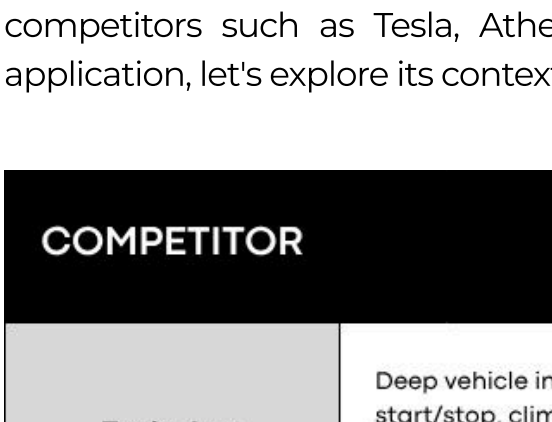
Instead of relying completely on a mobile app, the EV's built-in infotainment system itself becomes the main interface — showing things like:

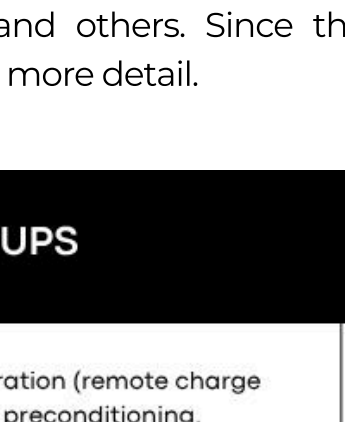
- Charging station navigation
- Battery health and range
- Service reminders
- Actionable battery health insights
- Trip planning

Basically, the EV app features live inside the car dashboard.

Advantages

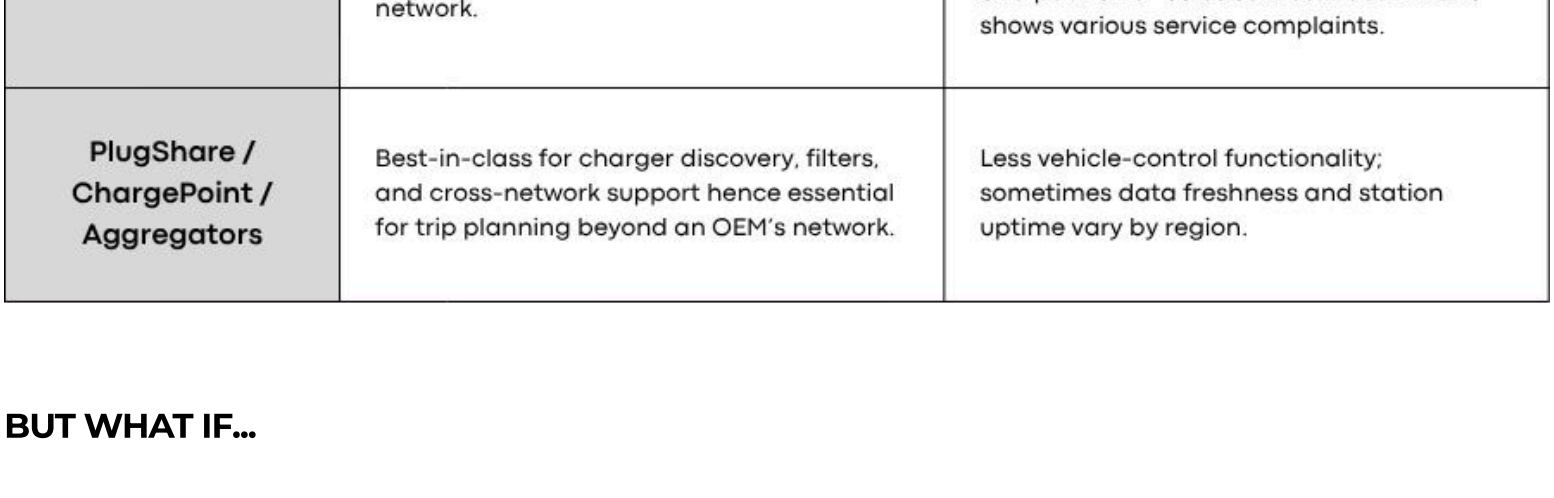
1. Seamless Experience: All information is visible where the driver needs it, reduces distraction since the driver doesn't need to look at the phone.
2. Larger & Contextual Display: Infotainment screens offer a bigger and clearer interface, ideal for maps, charger locations, and journey planning. It also allows intuitive controls through touch, steering buttons, or voice commands.

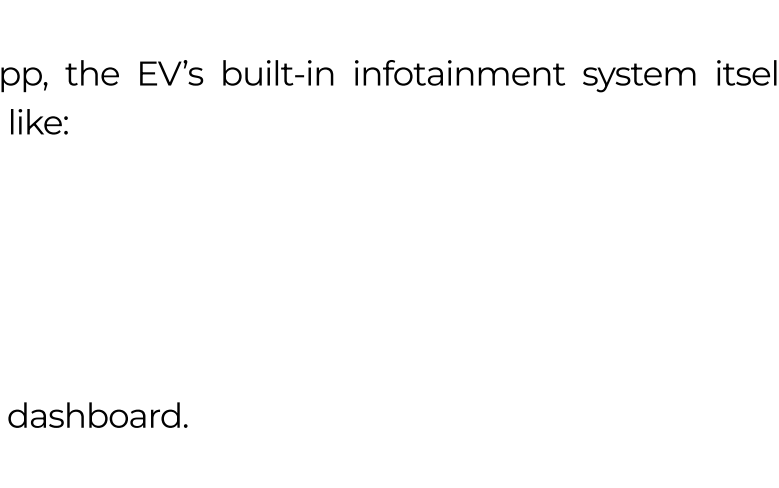




Challenges

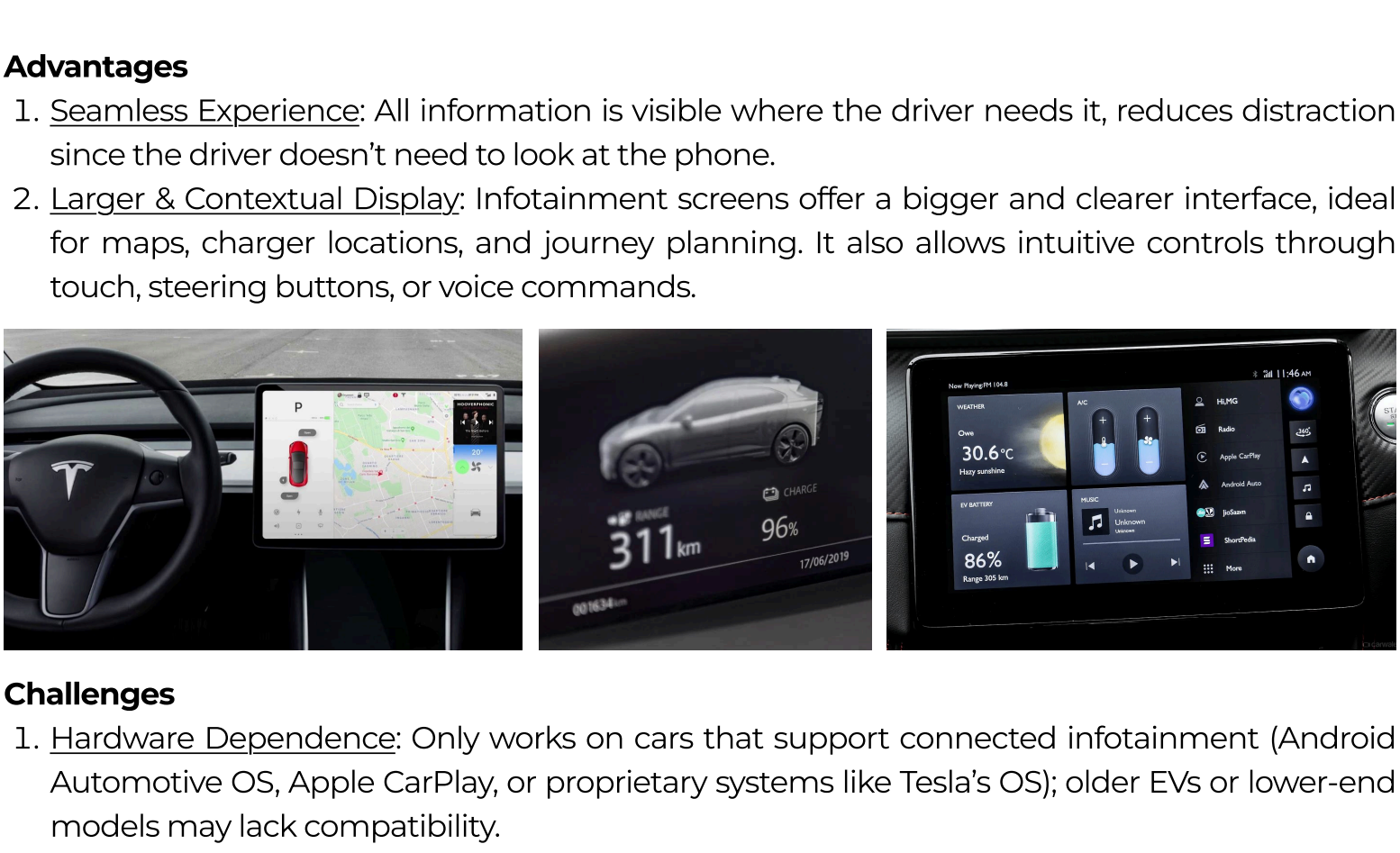
1. Hardware Dependence: Only works on cars that support connected infotainment (Android Automotive OS, Apple CarPlay, or proprietary systems like Tesla's OS); older EVs or lower-end models may lack compatibility.
2. Limited Third-Party Access: Manufacturers (like Tata, MG, or Ola) often restrict external apps from integrating deeply into the dashboard system.
3. Safety Regulations: Excessive touch interaction or animated visuals during driving can be illegal or unsafe.
4. Update Complexity: Updating or adding features requires firmware updates to the car system — not as easy as pushing app updates via Play Store/App Store.
5. Cost & Scalability: Building and certifying dashboard-integrated software is more expensive and slower than releasing a mobile app; for which a customized OS for each brand(Tesla OS, Android Automotive, etc.) will be required.

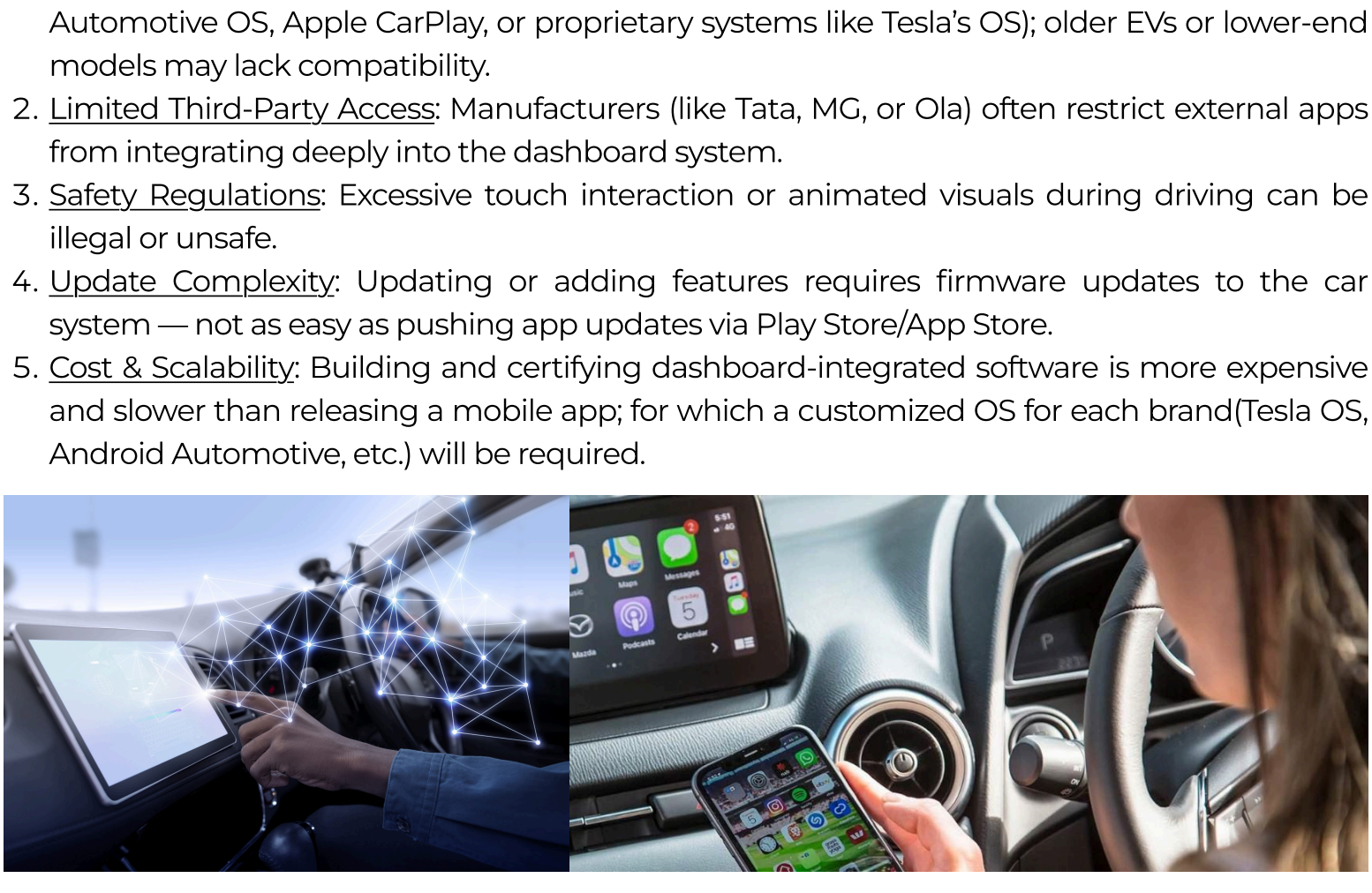




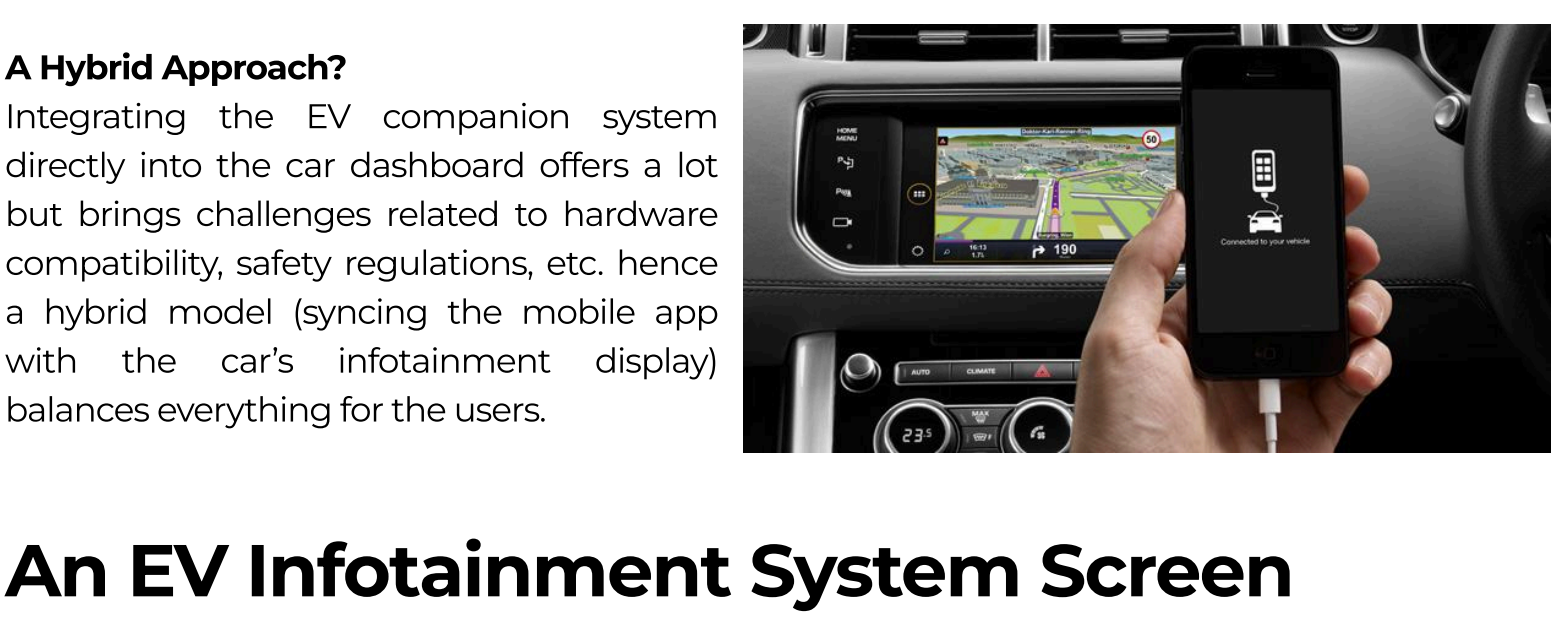
A Hybrid Approach?
Integrating the EV companion system directly into the car dashboard offers a lot but brings challenges related to hardware compatibility, safety regulations, etc. hence a hybrid model (syncing the mobile app with the car's infotainment display) balances everything for the users.

An EV Infotainment System Screen

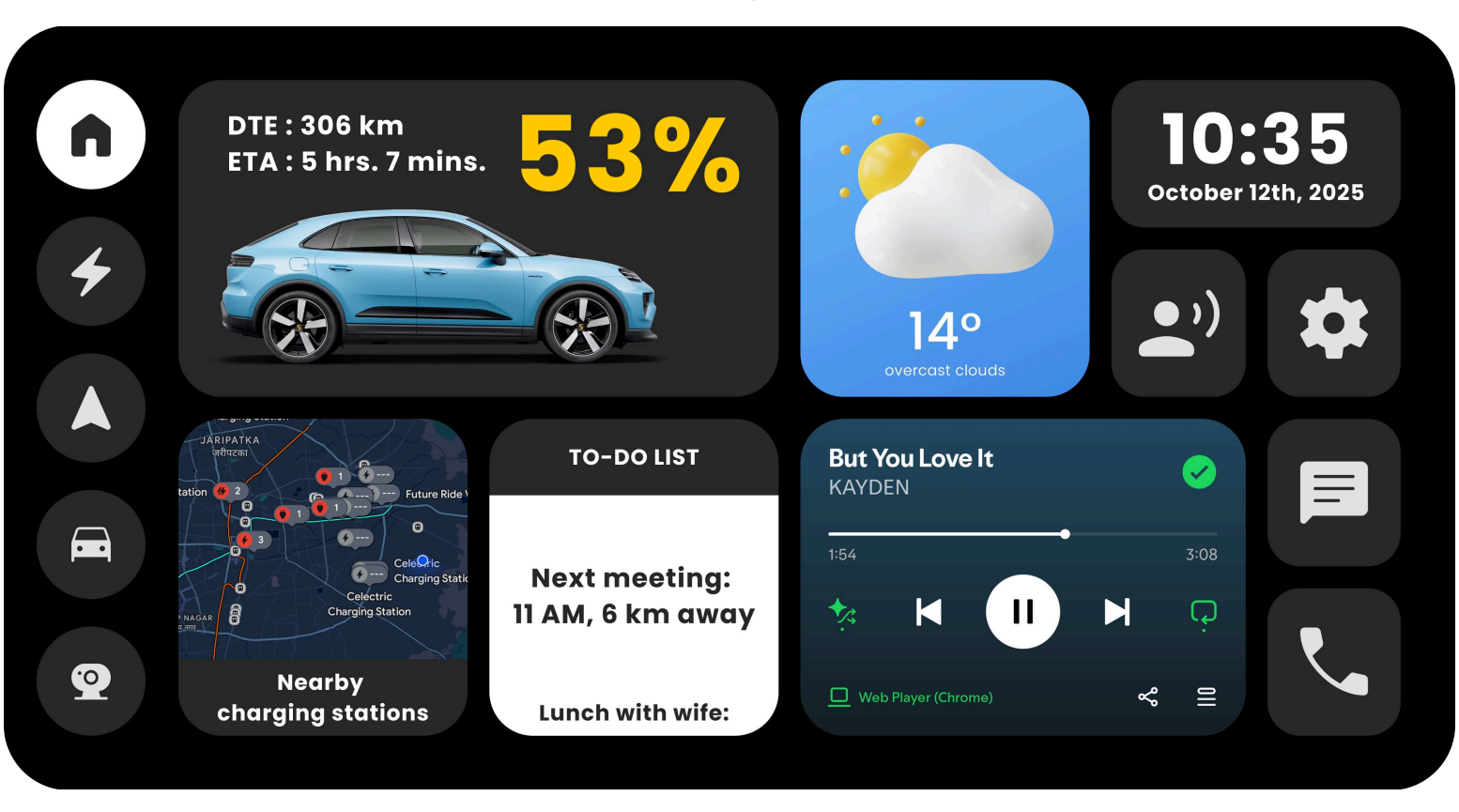




Average battery temp 38°C — avoid extended parking in direct sunlight.


INSIGHTS
10:35
October 12th, 2025
NEARBY CHARGING STATIONS
ABC Charging Station, ABCD
1.3 kms away | 3 mins
₹ 20/kWh – ₹800 full charge.
LET'S GO
XYZ Charging Station, XYZA
1.9 kms away | 5 mins
₹ 18/kWh – ₹720 full charge.
LET'S GO
ACTIVATE SMART NAVIGATION

The estimated cost for a full charge is approximately ₹390, based on your current charge level.



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